You

(Now Wednesday, Jun 11)
I'm making a long speedrun video on YouTube where the speedrun took me around 2 hours. I want to have a stopwatch on screen counting up the time, however in my video will be loads of cuts and I still want the stopwatch to progress with how much actual time it took. For example, I show the stopwatch for the first 5 mins then I cut to around 8 mins and show the stopwatch again at 8 mins. The thing is that I don't know when or where the cuts will be and I want to download a 2 hour video that counts up that I can then edit with my original recording. Can you point me to a 2-hour count-up video I can download for free from somewhere, or if that's not possible, how should I do it?

Copilot

You can download a **2-hour count-up timer video** for free from this YouTube video. It's a simple, no-sound timer that counts up continuously, which should work well for your speedrun editing.

If you prefer a more customized approach, you could create your own timer using **Adobe After Effects**. This tutorial walks you through setting up a count-up timer with keyframes, allowing you to control the timing precisely. That way, you can match the stopwatch to your actual speedrun duration without worrying about pre-made cuts.

Would you like help with integrating the timer into your video editing workflow?

You

I'm on the video you linked, but I can't download the count-up, I can only view it on Youtube, how do I download it?

Copilot

Downloading YouTube videos directly from the platform isn't officially supported unless you have **YouTube Premium**, which allows offline viewing within the app. However, there are alternative methods you can explore:

1. **YouTube Premium** – If you subscribe, you can download videos for offline viewing, but they remain within the YouTube app.
2. **Online Downloaders** – Websites like **4K Video Downloader** or **ClipGrab** allow you to save YouTube videos to your device tomshardware.com clipchamp.com.
3. **Browser Extensions** – Some Firefox and Chrome extensions can help download videos, though availability may vary clipchamp.com.
4. **Command-Line Tools** – If you're comfortable with tech, **youtube-dl** or **yt-dlp** are open-source tools that let you download videos tomshardware.com.

Keep in mind that downloading videos without permission may violate YouTube's terms of service. If you want a **custom stopwatch**, you could create one using **Adobe After Effects** or a simple Python script with OpenCV.

Would you like guidance on setting up a custom timer instead?

You

I get this message when I try to download it [Uploaded Image Expired], how can I make a script in Python that generates a video of a stopwatch counting up from a start time stamp that I enter to an end time stamp in hours, minutes and seconds and then lets me download/save the video as mp4 at a file path of my choice?

Copilot

You can create a Python script using **MoviePy** and **OpenCV** to generate a stopwatch video that counts up from a specified start time to an end time. Here's a complete script to achieve this:

Steps the script will follow:

1. Accept a **start timestamp** (hours, minutes, seconds).
2. Accept an **end timestamp** (hours, minutes, seconds).
3. Generate frames displaying the current elapsed time.
4. Save the final video as an MP4 file at the desired location.

Python Script:

```
import cv2
import numpy as np
from moviepy.editor import VideoClip

def generate_stopwatch_video(start_time, end_time, file_path):
    start_seconds = start_time[0] * 3600 + start_time[1] * 60 + start_time[2]
    end_seconds = end_time[0] * 3600 + end_time[1] * 60 + end_time[2]
    duration = end_seconds - start_seconds

    def make_frame(t):
        total_seconds = int(t) + start_seconds
        hours, remainder = divmod(total_seconds, 3600)
        minutes, seconds = divmod(remainder, 60)
        time_str = f"{hours:02}:{minutes:02}:{seconds:02}"

        frame = np.zeros((200, 400, 3), dtype=np.uint8)  # Black background
        cv2.putText(frame, time_str, (50, 100), cv2.FONT_HERSHEY_SIMPLEX, 2, (255, 255, 255), 3, cv2.LINE_AA)
        return frame

    video = VideoClip(make_frame, duration=duration)
    video.write_videofile(file_path, fps=24)

# Example usage:
start_time = (0, 0, 0)  # Start at 00:00:00
end_time = (2, 0, 0)    # End at 02:00:00
file_path = "stopwatch_video.mp4"

generate_stopwatch_video(start_time, end_time, file_path)
```

Explanation:

- The script calculates the total elapsed **seconds** from the provided timestamps.
- It creates video frames dynamically using OpenCV.
- The stopwatch text is updated per frame and embedded on a blank video background.
- The **final video** is exported as an MP4 file with a **24 FPS** frame rate.

This method ensures that the stopwatch correctly follows your intended speedrun timing even with cuts. Would you like additional features, such as **custom colors or fonts** for the stopwatch display?

You

what would be the pip install commands for MoviePy and OpenCV

Copilot

You can install **MoviePy** and **OpenCV** using the following commands:
pip install moviepy
pip install opencv-python
If you also need OpenCV's **full package** with extra functionalities, use:
pip install opencv-python-headless
This version is useful if you **don't** need OpenCV's GUI features (e.g., window display functions) and just want the core processing capabilities.
Let me know if you run into any installation issues! 🚀

You

Traceback (most recent call last):
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\video\VideoClip.py", line 342, in write_videofile
codec = extensions_dict[ext]["codec"][0]
*~~~~~~~~~~~~~~~~^^^^*
KeyError: ''
During handling of the above exception, another exception occurred:
Traceback (most recent call last):
File "c:\Users\fifau\Downloads\generate_time_video.py", line 35, in <module>
generate_stopwatch_video(start_time, end_time, file_path)
File "c:\Users\fifau\Downloads\generate_time_video.py", line 21, in generate_stopwatch_video
video.write_videofile(file_path, fps=30)
File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
*return caller(func, *(extras + args), **kw)*
*^^*
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py", line 53, in requires_duration
*return func(clip, *args, **kwargs)*
*^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^*
File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
*return caller(func, *(extras + args), **kw)*
*^^*
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py", line 153, in wrapper
*return func(clip, *new_args, **new_kwargs)*
*^^*
File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
*return caller(func, *(extras + args), **kw)*
*^^*
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py", line 24, in convert_masks_to_RGB
*return func(clip, *args, **kwargs)*
*^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^*
File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
*return caller(func, *(extras + args), **kw)*
*^^*
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py", line 102, in wrapper
*return func(*new_args, **new_kwargs)*
*^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^*
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\video\VideoClip.py", line 344, in write_videofile
raise ValueError(
ValueError: MoviePy couldn't find the codec associated with the filename. Provide the 'codec' parameter in write_videofile.

Copilot

It looks like **MoviePy** is unable to determine the correct codec for encoding your MP4 file. You can resolve this issue by explicitly specifying the codec when calling write_videofile.
Try this updated line in your script:

video.write_videofile(file_path, fps=30, codec="libx264")

Why this happens:

- MoviePy tries to determine the codec based on your file extension, but sometimes it fails.
- "libx264" is a commonly used codec for MP4 videos and should work.
- Some systems might require **FFmpeg** installed for proper video encoding.

If you still encounter issues, try installing or updating **FFmpeg** using:

pip install imageio[ffmpeg]

Let me know if this solves the problem or if the error persists!

You

Enter the hour of the starting time (Press Enter if 0): 0
Enter the minute of the starting time: 0
Enter the second of the starting time: 0
Enter the hour of the ending time (Press Enter if 0): 0
Enter the minute of the ending time: 1
Enter the second of the ending time: 36
Enter the filepath for where to save it: C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds

```
MoviePy - Building video C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds.
MoviePy - Writing video C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds
frame_index:   0%|                                        | 2/2880 [00:00<03:56, 12.15it/s, now=None]
Traceback (most recent call last):
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-
packages\moviepy\video\io\ffmpeg_writer.py", line 169, in write_frame
    self.proc.stdin.write(img_array.tobytes())
BrokenPipeError: [Errno 32] Broken pipe

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "c:\Users\fifau\Downloads\generate_time_video.py", line 35, in <module>
    generate_stopwatch_video(start_time, end_time, file_path)
  File "c:\Users\fifau\Downloads\generate_time_video.py", line 21, in generate_stopwatch_video
    video.write_videofile(file_path, fps=30, codec="libx264")
  File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
    return caller(func, *(extras + args), **kw)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py",
line 53, in requires_duration
    return func(clip, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
    return caller(func, *(extras + args), **kw)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py",
line 153, in wrapper
    return func(clip, *new_args, **new_kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
    return caller(func, *(extras + args), **kw)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py",
line 24, in convert_masks_to_RGB
    return func(clip, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Roaming\Python\Python312\site-packages\decorator.py", line 232, in fun
    return caller(func, *(extras + args), **kw)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\decorators.py",
line 102, in wrapper
    return func(*new_args, **new_kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-
packages\moviepy\video\VideoClip.py", line 393, in write_videofile
```

ffmpeg_write_video(
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\video\io\ffmpeg_writer.py", line 295, in ffmpeg_write_video
writer.write_frame(frame)
File "C:\Users\fifau\AppData\Local\Programs\Python\Python312\Lib\site-packages\moviepy\video\io\ffmpeg_writer.py", line 221, in write_frame
raise IOError(error)
OSError: [Errno 32] Broken pipe

MoviePy error: FFMPEG encountered the following error while writing file C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds:

[AVFormatContext @ 00000225bbf10840] Unable to choose an output format for 'C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds'; use a standard extension for the filename or specify the format manually.
[out#0 @ 00000225bbec3b00] Error initializing the muxer for C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds: Invalid argument
Error opening output file C:\Users\fifau\Videos\Screen Recordings\Memes-Effects-Sounds.
Error opening output files: Invalid argument

Copilot

You're encountering a **MoviePy FFMPEG error** because the filename you provided **does not have a file extension**.
Fix: Add .mp4 to the filename
Modify the file path like this:

```
file_path = "C:\\Users\\fifau\\Videos\\Screen Recordings\\Memes-Effects-Sounds.mp4"
```

Then rerun your script with this proper **MP4 file extension**.
Why this happens?

- MoviePy relies on the file extension (.mp4, .avi, .mov) to determine the **correct output format**.
- If no extension is provided, **FFMPEG fails to choose a format** and throws an error.

Once you've updated the filename, give it another shot and let me know how it goes! 🚀

Conversation by Fifa Utuber with Copilot on Wednesday, Jun 11 2025